Informatik
Exercise Session

Consider the normalized floating point number system $F^*(\beta, p, e_{\min}, e_{\max})$ with $\beta = 2$, $p = 3$, $e_{\min} = -4$, $e_{\max} = 4$.
Compute the following expressions as the parentheses suggest, representing each intermediate result (and the final result) in the normalized floating point system according to the rules of computing with floating point numbers.

$(10 + 0.5) + 0.5$

	decimal		binary
	10		?????
+	0.5		?????
=			?????
+	0.5		?????
=	??	←	?????

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		?????
+	0.5		?????
=			?????
+	10		?????
=	??	←	?????

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			?????
$+$	0.5		?????
$=$	??	$\leftarrow$	?????

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		?????
$+$	0.5		?????
$=$			?????
$+$	10		?????
$=$	??	$\leftarrow$	?????

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.0101 \cdot 2^3$
$+$	0.5		?????
$=$	??	$\leftarrow$	?????

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		?????
$+$	0.5		?????
$=$			?????
$+$	10		?????
$=$	??	$\leftarrow$	?????

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$	??	$\leftarrow$	?????

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		?????
$+$	0.5		?????
$=$			?????
$+$	10		?????
$=$	??	$\leftarrow$	?????

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$	10	$\leftarrow$	$1.01 \cdot 2^3$

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		?????
$+$	0.5		?????
$=$			?????
$+$	10		?????
$=$	??	$\leftarrow$	?????

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$	10	$\leftarrow$	$1.01 \cdot 2^3$

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		$1.00 \cdot 2^{-1}$
$+$	0.5		$1.00 \cdot 2^{-1}$
$=$			?????
$+$	10		?????
$=$	??	$\leftarrow$	?????

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$	10	$\leftarrow$	$1.01 \cdot 2^3$

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		$1.00 \cdot 2^{-1}$
$+$	0.5		$1.00 \cdot 2^{-1}$
$=$			$1.00 \cdot 2^0$
$+$	10		$1010.00 \cdot 2^0$
$=$	??	$\leftarrow$	?????

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$	10	$\leftarrow$	$1.01 \cdot 2^3$

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		$1.00 \cdot 2^{-1}$
$+$	0.5		$1.00 \cdot 2^{-1}$
$=$			$1.00 \cdot 2^0$
$+$	10		$1010.00 \cdot 2^0$
$=$	??	$\leftarrow$	$1011.00 \cdot 2^0$

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$	10	$\leftarrow$	$1.01 \cdot 2^3$

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		$1.00 \cdot 2^{-1}$
$+$	0.5		$1.00 \cdot 2^{-1}$
$=$			$1.00 \cdot 2^0$
$+$	10		$1010.00 \cdot 2^0$
$=$	??	$\leftarrow$	$1.011 \cdot 2^3$

$(10 + 0.5) + 0.5$

	decimal		binary
	10		$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$			$1.01 \cdot 2^3$
$+$	0.5		$0.0001 \cdot 2^3$
$=$	10	$\leftarrow$	$1.01 \cdot 2^3$

$(0.5 + 0.5) + 10$

	decimal		binary
	0.5		$1.00 \cdot 2^{-1}$
$+$	0.5		$1.00 \cdot 2^{-1}$
$=$			$1.00 \cdot 2^0$
$+$	10		$1010.00 \cdot 2^0$
$=$	12	$\leftarrow$	$1.10 \cdot 2^3$